4. Monitoring and alerting
5. Operating

Although you probably have some idea of what these things mean, let's describe them so that we're on the same page.

Provisioning—also known as creating stuff. Software needs computers and resources to run on. Ensuring there are enough computing resources available—and creating them if not—is an important part of that. In fact, it's not just *machines* or *instances* that we need; when deploying apps to the cloud, we may also need other physical or virtual infrastructure, such as:

- Networks, also known as virtual private clouds (VPCs)
- Network address translation (NAT) gateways
- Routers
- Firewalls
- Internet gateways
- Proxies
- Security rules

Configuration management. Once our raw infrastructure has been created, it typically isn't ready to perform its true purpose in life; it has to be configured. For some (possibly virtual) infrastructure, such as routers or firewalls, this means ensuring they have the right settings or rules applied. For a server, it typically means installing the necessary software packages and dependencies needed to run an application.

Release management. Most systems need regular maintenance or continued development, which implies an ongoing need to deploy—or *release*—newer versions. Typically, we'd like to be able to release easily, repeatedly at any time, and with minimal or no impact to users of the system. This can be achieved through automation, good tooling, and techniques such as *blue-green deploys* or *canary releases*. Additionally, when things go wrong, the ability to roll back to a known-good version of the software can be invaluable.

Monitoring and alerting. In a perfect world, our application, once deployed, would perform as expected forever. In reality, things go wrong: servers fail, software has bugs, our assumptions don't hold true. Rather than bury our heads in the sand, we proactively take steps to know the health of our software and how well it's functioning. This typically involves tracking health metrics for the system and being alerted when things start to go wrong.

Operating. In the day-to-day running of production systems, there are some common things we may need to do: scaling up to handle increased load,